

COMPARATIVE ANALYSIS OF SARIMA-GARCH AND LSTM FOR SOIL MOISTURE FORECASTING IN INDRAMAYU

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ABSTRACT

Soil moisture is a critical determinant of agricultural productivity and hydrological stability, particularly in Indramayu Regency, a major rice-producing region in West Java vulnerable to seasonal drought. Accurate forecasting is essential for optimizing irrigation management and mitigating climate risks. This study evaluates and compares the performance of a hybrid statistical model, SARIMA-GARCH, against a Deep Learning approach using Long Short-Term Memory (LSTM) networks to forecast soil moisture. The study utilizes weekly data from 2014 to 2023, employing SARIMA-GARCH to address seasonality and heteroskedasticity, while LSTM is optimized to capture non-linear temporal dependencies. The evaluation results on 2023 test data demonstrate that LSTM significantly outperforms the statistical hybrid model. The LSTM model achieved a Root Mean Square Error (RMSE) of 0.0157 and a Mean Absolute Percentage Error (MAPE) of 3.52%, compared to 0.0399 and 9.94% for SARIMA-GARCH. Visually, LSTM produced a smooth, realistic trajectory aligned with physical soil inertia, whereas SARIMA-GARCH exhibited excessive noise. Consequently, this study establishes LSTM as the superior architecture for precise soil moisture forecasting to support regional food security policy.

Keywords: Soil Moisture, SARIMA-GARCH, Long Short-Term Memory (LSTM), Indramayu Regency.

INTRODUCTION

Soil moisture refers to the amount of water stored within the soil pores, either volumetrically or gravimetrically, and is often used as an indicator of a region's hydrological conditions (Ruszczak & Boguszewska-Mańkowska, 2022). Adequate soil moisture is crucial because it can enhance soil fertility and support plant growth (Yang et al., 2022). Conversely, poor soil moisture conditions may cause problems such as reduced root water uptake or increased risks of erosion and nutrient loss (Chen & Wu, 2024). Monitoring and understanding soil moisture is therefore essential, as it influences the water cycle, irrigation management, and adaptation to climate change (Rasheed et al., 2022).

Poorly managed soil moisture leads to increased surface runoff and topsoil erosion, which subsequently affects water quality and agricultural land productivity (DeVilleneuve et al., 2023). These conditions arise from soil textures that do not support adequate water retention, land-use changes, and inefficient irrigation practices (Engels et al., 2025). Moreover, fluctuations in soil moisture extending into the root zone can influence crop yields and even future food security (Noory et al., 2025), highlighting the importance of having a soil-moisture prediction system as part of agronomic and hydrological management. Therefore, modelling and forecasting soil moisture serve as essential components in mitigating agronomic and environmental risks (Shawon et al., 2025).

Soil moisture conditions play a crucial role in regulating irrigation and determining rice productivity (Liu et al., 2025), especially in regions with extensive paddy fields such as Indramayu Regency. Indramayu possesses the largest rice-field area in West Java, covering 115,897 hectares, or 55.20% of the regency's total land area (Muttaqien et al., 2020). Due to its relatively stable geographic characteristics and seasonal patterns, Indramayu is considered an appropriate site for soil-moisture

analysis and forecasting (Nanda, 2025). Consequently, focusing on Indramayu Regency provides both scientific value and practical benefits for local policymaking.

Despite this strategic importance, Indramayu is highly vulnerable to seasonal drought and climate change impacts (Perdinan et al., 2014), which pose a continuous threat to rice production. Studies, such as that conducted by Darojati et al. (2015), underscore the urgency of drought hazard monitoring and the need for effective mitigation strategies in the region. The high accuracy of soil moisture forecasting models, therefore, offers a crucial tool for irrigation planning, informing water policy, and mitigating crop loss, thereby contributing directly to regional food security (Zhang et al., 2025).

This study employs the Seasonal Autoregressive Integrated Moving Average (SARIMA) model to forecast soil moisture in Indramayu Regency for the year 2024 using data from 2014–2023. SARIMA was selected due to its ability to capture seasonal patterns and trends in time-series data, making it suitable for soil-moisture data that exhibit periodic cycles (Kumari & Muthulakshmi, 2024). Comparing forecast results with actual observations provides a concrete evaluation of the model's accuracy and helps assess whether SARIMA is sufficiently reliable to support irrigation policy and agronomic risk-mitigation strategies (Szostek et al., 2024).

However, in practice, soil-moisture data often exhibit volatility due to rainfall anomalies. Previous studies (Isuari et al., 2022) have demonstrated that standard SARIMA models struggle to capture such extreme fluctuations or structural breaks without additional intervention or volatility modeling. Therefore, this study also integrates the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model into SARIMA to form the combined SARIMA–GARCH model, which detect and model residual heteroskedasticity. In addition to the statistical approach, this research compares the performance of SARIMA–GARCH with the Long Short-Term Memory (LSTM) model, a neural-network architecture that excels in learning sequential patterns and capturing long-term dependencies in nonlinear time-series data. This comparison aims to determine which model provides the most accurate forecasting performance.

METHOD

Data

The data used in this study consist of daily soil-moisture observations from the ESA CCI Soil Moisture Dataset, accessed via the CEDA Archive. The dataset is a cross-calibrated merge of multiple active and passive satellite sensors producing global soil-moisture estimates (m^3/m^3) since 1978 (Gruber et al., 2019). Daily NetCDF (.nc) files for each year were downloaded and processed in Python using xarray to open and merge the data. The analysis focuses on the period 1 January 2014–31 December 2023, which was aggregated into weekly averages, resulting in 522 weekly observations. The spatial extraction was limited to Indramayu Regency, West Java (6.25° – 6.67° S; 107.87° – 108.60° E), with soil moisture values averaged across all pixels in this area to obtain a single representative value per time point.

SARIMA-GARCH Methodology

The initial stage of this research involved exploratory data analysis using time series plots, histograms, and boxplots to examine the behaviour of the soil-moisture series. These visualizations helped to identify potential outliers, shifts in variance, and distributional characteristics that could influence the subsequent modelling steps. Insights from this exploratory phase also guided decisions regarding the appropriate data-splitting strategy for model evaluation, which was determined after assessing the temporal structure and stability of the series. The results of this stage further informed whether transformation or differencing was required before proceeding to model identification.

A preliminary transformation analysis was conducted to ensure that the series met the assumptions required for Seasonal Autoregressive Integrated Moving Average (SARIMA) modelling. The Box–Cox

plot was examined to detect variance instability and non-normality. When the estimated λ (lambda) value suggested the need for variance stabilization, the Box–Cox power transformation was applied as follows:

$$Y_t^{(\lambda)} = \begin{cases} \frac{Y_t^\lambda - 1}{\lambda}, & \lambda \neq 0 \\ \ln(Y_t), & \lambda = 0 \end{cases} \quad (1)$$

In addition to transformation, the autocorrelation structure of the series was analyzed using the Autocorrelation Function (ACF) plot to detect persistence, seasonal correlation, and potential non-stationarity. A formal assessment of unit-root behavior was conducted using the Augmented Dickey–Fuller (ADF) test, evaluating the hypothesis:

H_0 : The series has a unit root (non-stationary)

H_1 : The series is stationary

The ADF test estimates the following regression model:

$$\Delta y_t = \alpha + \beta t + \gamma y_{t-1} + \sum_{i=1}^k \delta_i \Delta y_{t-i} + \varepsilon_t \quad (2)$$

Beyond the formal test, the ACF plot plays a crucial role in diagnosing stationarity. For non-seasonal components, a slow and gradual decay in the ACF typically signals non-stationarity. For seasonal series, strong spikes at multiples of the seasonal period (e.g., lag s , $2s$, ...) accompanied by persistent correlation indicate seasonal non-stationarity. When these patterns appear, non-seasonal differencing $(1 - B)$ or seasonal differencing $(1 - B^s)$ is generally required. When non-stationarity persisted, differencing was applied to remove deterministic or seasonal trends. The general form of ordinary differencing is given by:

Ordinary Differencing (Order d)

$$\nabla^d Y_t = (1 - B)^d Y_t \quad (3)$$

Seasonal Differencing (Order D , Seasonal Period s)

$$\nabla_s^D Y_t = (1 - B^s)^D Y_t \quad (4)$$

Seasonal differencing was specifically applied when significant autocorrelation appeared at seasonal lags. Following this transformation, the adequacy of the step was re-evaluated using ACF and ADF analysis. This verification is crucial to confirm both variance stability and stationarity before proceeding to SARIMA model identification.

The identification of the SARIMA structure begins by determining the appropriate non-seasonal (p, d, q) and seasonal $(P, D, Q)_s$ orders through patterns observed in the ACF, Partial Autocorrelation Function (PACF), and Extended Autocorrelation Function (EACF) of the stationarized series. This structure incorporates both differencing operators and autoregressive–moving average components at seasonal and non-seasonal levels. The general form of the SARIMA(p, d, q)(P, D, Q) $_s$ that follows the standard representation described by (Wilson, 2016) model can be expressed as:

$$\Phi_P(B^s) \phi_p(B) (1 - B)^d (1 - B^s)^D y_t = \Theta_Q(B^s) \theta_q(B) \varepsilon_t \quad (5)$$

With:

$\phi_p(B)$: non-seasonal autoregressive (AR) polynomial of order p

$\theta_q(B)$: non-seasonal moving average (MA) polynomial of order q

$\Phi_P(B^s)$: seasonal AR polynomial of order P with seasonal lag s

$\Theta_Q(B^s)$: seasonal MA polynomial of order Q with seasonal lag s

$(1 - B)^d$: non-seasonal differencing of order d

$(1 - B^s)^D$: seasonal differencing of order D

ε_t : white noise error term

Model candidates are then estimated and compared using information criteria, where the optimal model is selected based on the lowest Akaike Information Criterion (AIC) and Bayesian Information Criterion (BIC) values. These are defined as:

$$AIC = -2 \ln(L) + 2k \quad (6)$$

$$BIC = -2 \ln(L) + k \ln(n) \quad (7)$$

A series of diagnostic tests was conducted to evaluate whether the fitted model satisfied the assumptions required for valid statistical inference. These diagnostics assessed the normality, independence, homoscedasticity, and zero-mean properties of the model residuals. The procedures and corresponding formulations are summarized as follows.

The normality of the residuals was assessed using the Shapiro–Wilk test, which is widely regarded as one of the most powerful tests for detecting deviations from normality in small to medium sample sizes. The general form of the Shapiro–Wilk test statistic is given by:

$$W = \frac{(\sum_{i=1}^n a_i x_{(n+1-i)} - a_i x_{(i)})^2}{\sum_{i=1}^n (x_i - \bar{x})^2} \quad (8)$$

The hypotheses are:

H_0 : Residuals follow a normal distribution

H_1 : Residuals deviate from normality

To further assess whether the residuals behave as white noise, the Ljung–Box test was applied to evaluate the presence of autocorrelation across multiple lags. The Ljung–Box statistic is defined as:

$$Q_{LB} = n(n+2) \sum_{k=1}^h \frac{\widehat{\rho_k^2}}{n-k} \quad (9)$$

The hypotheses for this test are:

H_0 : No autocorrelation in residuals

H_1 : Autocorrelation in residuals

To test for homogeneity (constant variance), the Ljung–Box test was also applied to the squared residuals (ε_t^2), which allows detection of ARCH effects (Sirius et al., 2023). The test statistic for squared residuals is:

$$Q_{LB} = n(n+2) \sum_{k=1}^m \frac{\widehat{\rho_{k,\varepsilon^2}^2}}{n-k} \quad (10)$$

The hypotheses for testing constant variance are:

$H_0: \rho_{1,\varepsilon^2} = \rho_{2,\varepsilon^2} = \dots = \rho_{m,\varepsilon^2} = 0$ (constant variance)

H_1 : There exists $k \in \{1, 2, \dots, m\}$ such that $\rho_{k,\varepsilon^2} \neq 0$ (non-constant variance)

The zero-mean assumption of the residuals was examined using the one-sample t -test. This test is particularly appropriate because it accounts for sampling variability when evaluating whether the average residual systematically departs from zero (Mircioiu & Atkinson, 2017). The general form of the t -statistic is expressed as:

$$t = \frac{\bar{e} - 0}{s_e / \sqrt{n}}, \quad (11)$$

The hypotheses are for t -statistic:

$H_0: \mu_e = 0$ residuals have zero means

$H_1: \mu_e \neq 0$ residuals deviate from zero mean

An overfitting assessment was carried out after obtaining the tentative SARIMA model to confirm its robustness. A more complex model was estimated by increasing some of the AR, MA, and seasonal parameters in order to evaluate whether the additional complexity could enhance diagnostic results or forecasting accuracy. This procedure served as a comparative check to determine whether the tentative model was truly the most appropriate specification. When the overfitted model did not provide meaningful improvement, the simpler tentative model was retained in accordance with the principle of parsimony (Box et al., 2015).

The analysis of volatility began by applying the ARCH Lagrange Multiplier (LM) test to the residuals of the SARIMA model. The test was conducted using the auxiliary regression of squared residuals, expressed as:

$$\varepsilon_t^2 = \gamma_0 + \gamma_1 \varepsilon_{t-1}^2 + \gamma_2 \varepsilon_{t-2}^2 + \dots + \gamma_q \varepsilon_{t-q}^2 + u_t \quad (12)$$

A significant test statistic indicated the presence of ARCH effects, which justified the use of a conditional variance model. After confirming heteroskedasticity, the Generalized Autoregressive Conditional Heteroskedasticity (GARCH) model was estimated. The general form of the GARCH(r, s) model considered in this study is:

$$\sigma_t^2 = \omega + \sum_{i=1}^r \alpha_i \varepsilon_{t-i}^2 + \sum_{j=1}^s \beta_j \sigma_{t-j}^2 \quad (13)$$

With:

σ_t^2 : Conditional variance at time t

ω : Constant term

ε_{t-i}^2 : Lagged squared residuals (ARCH terms)

α_i : ARCH parameters

σ_{t-j}^2 : Lagged conditional variances (GARCH terms)

β_j : GARCH parameters
 r, s : Order of the GARCH model

Where α_i captures the effect of past shocks and β_j represents the persistence of past volatility. The parameter search was performed systematically, beginning with the simplest structure GARCH(1,1) and extending to GARCH(2,2). For each specification, two distributional assumptions were evaluated: the normal distribution and the Student t distribution.

Once the diagnostic criteria are satisfied, multi-step volatility forecasting is generated from the fitted GARCH(r, s) specification. In parallel, point forecasts of the conditional mean are obtained from the selected SARIMA(p, d, q)(P, D, Q)_s model. The SARIMA forecasting equation can be written as:

$$\widehat{y}_{t+h} = \mu + \sum_{i=1}^p \phi_i \widehat{y}_{t+h-i} + \sum_{j=1}^q \theta_j \widehat{\varepsilon}_{t+h-j} + \sum_{k=1}^P \Phi_k \widehat{y}_{t+h-ks} + \sum_{l=1}^Q \Theta_l \widehat{\varepsilon}_{t+h-ls} \quad (14)$$

The final forecast is subsequently formed by combining the SARIMA mean forecast with the GARCH-based volatility adjustment, producing:

$$\widehat{Y}_{t+h}^{\text{final}} = \widehat{y}_{t+h} + \widehat{\sigma}_{t+h} \quad (15)$$

LSTM Methodology

Before being fed into the Long Short-Term Memory (LSTM) architecture, the time series is normalized using the StandardScaler transformation to stabilize variance and improve training efficiency. This transformation rescales the data to have zero mean and unit variance, ensuring that the optimization process is not biased by differences in magnitude across observations. The normalization is defined as:

$$z_t = \frac{x_t - \mu}{\sigma} \quad (16)$$

As a comparative artificial intelligence-based method, this study employs a LSTM model designed to capture temporal dependencies through a set of internal gating mechanisms. After normalization, the time series is transformed into supervised learning format using a sliding-window approach, enabling the model to learn patterns from fixed-length historical sequences. The internal dynamics of the LSTM with ReLU activation for both the candidate state and hidden-state output can be expressed as:

$$f_t = \sigma(W_f x_t + U_f h_{t-1} + b_f) \quad (17)$$

$$i_t = \sigma(W_i x_t + U_i h_{t-1} + b_i) \quad (18)$$

$$\tilde{c}_t = \text{ReLU}(W_c x_t + U_c h_{t-1} + b_c) \quad (19)$$

$$c_t = f_t \odot c_{t-1} + i_t \odot \tilde{c}_t \quad (20)$$

$$o_t = \sigma(W_o x_t + U_o h_{t-1} + b_o) \quad (21)$$

$$h_t = o_t \odot \text{ReLU}(c_t) \quad (22)$$

x_t : The input vector at time t.

h_{t-1} : the short-term memory that the model carries forward

i_t : Forget Gate Activation

$\tilde{c}_t$: Input Gate Activation

c_t : Cell State (Long-Term Memory)

o_t : Output Gate Activation
 h_t : Hidden State (Short-Term Memory)
 W_f, W_i, W_c, W_o : Weight Matrices
 U_f, U_i, U_c, U_o : Recurrent Weight Matrices
 b_f, b_i, b_c, b_o : Bias Terms

To ensure that the LSTM model is configured optimally for the characteristics of the time series data, a structured hyperparameter tuning procedure is implemented. The model is trained under multiple architectural and training configurations to evaluate its sensitivity to different settings and to identify the combination that yields the most stable and accurate forecasting performance. The key characteristics and specifications considered during this tuning process are summarized in Table 1.

Table 1. Characteristics and Specifications of the LSTM Model with Manual Grid Search

Characteristic	Specification
Architecture	Seq2Seq (1×64 Encoder, 1×64 Decoder)
Time Steps	52
Epoch	50 & 100
Batch Size	16, 32, 64
Drop Out	0.1 & 0.05
Optimizer	ADAM

This study employs a Sequence-to-Sequence (Seq2Seq) LSTM architecture with single Encoder and Decoder layers (64 neurons each) and a 52-week window size to align with SARIMA's seasonality. The model is optimised using ADAM, incorporating dropout rates of 0.05 and 0.1 to prevent overfitting. To determine the optimal configuration, a grid search is conducted across batch sizes (16, 32, 64) and training epochs (50, 100).

Model Evaluation

The final stage of the study involved assessing the forecasting performance of the SARIMA–GARCH and LSTM models. In this comparison, two evaluation metrics were utilized: the Root Mean Squared Error (RMSE) and the Mean Absolute Percentage Error (MAPE). These metrics were chosen because they capture different aspects of prediction accuracy, with lower values indicating better model performance. The evaluation metrics are defined as follows:

$$\text{RMSE} = \sqrt{\frac{1}{n} \sum_{t=1}^n (y_t - \hat{y}_t)^2} \quad (23)$$

The RMSE measures the magnitude of prediction deviations from the actual values, applying a heavier penalty to larger errors and therefore remaining sensitive to outliers. According to (Hewamalage et al., 2023), RMSE is widely used as it provides a direct indication of model accuracy in the same units as the original data.

$$\text{MAPE} = \frac{1}{n} \sum_{t=1}^n \left| \frac{Y_t - \hat{Y}_t}{Y_t} \right| \times 100\% \quad (24)$$

The MAPE quantifies the average absolute prediction error as a percentage of the actual values, making it intuitive and easy to interpret. A study by (Buturac, 2022) highlights that MAPE is commonly employed for evaluating time-series forecasting performance because it is scale-independent.

RESULTS AND DISCUSSION

Data Exploration

To provide a clearer understanding of the underlying patterns in the soil moisture data for Indramayu Regency, two visual analyses are presented. The first visualization illustrates the division of the dataset into training and testing segments, while the second presents the time series decomposition to reveal the structural components that inform subsequent methodological decisions.

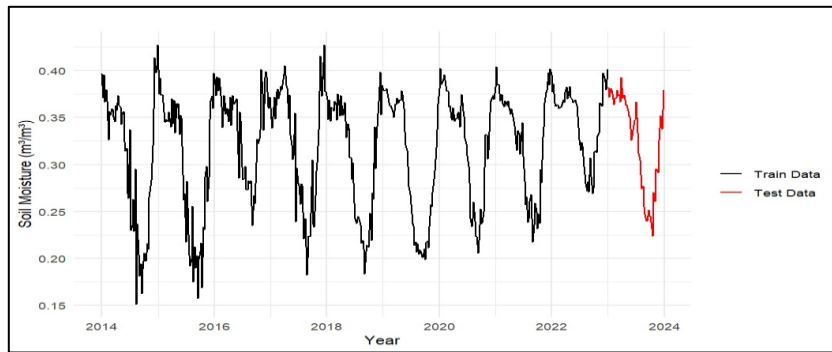


Figure 1. Soil Moisture Data Exploration: Plot of training and testing data split

Figure 1 displays the soil moisture time series divided using a 90:10 hold-out scheme. The data from January 2014 to December 2022 are allocated as the training set, whereas the period from January 2023 to December 2023 serves as the testing set. This separation provides a clear temporal boundary between the model development phase and the model evaluation phase, ensuring that forecasting accuracy is assessed on unseen data.

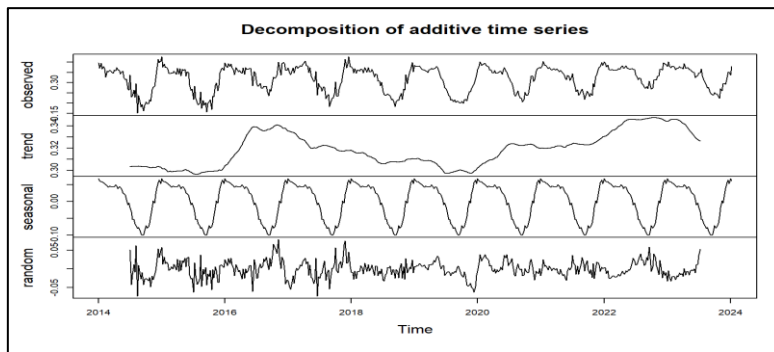


Figure 2. Soil Moisture Data Exploration: Time series component decomposition.

Figure 2 presents the decomposition of the soil moisture series into its trend, seasonal, and residual components. The trend component demonstrates nonlinear movements, indicating non-stationarity in the mean and the necessity of differencing. The seasonal component shows a recurring annual hydrological pattern, where soil moisture increases during the rainy season and declines during

the dry season, justifying the use of seasonal orders in the SARIMA framework. Meanwhile, the residual component exhibits variability that is not constant over time, suggesting the presence of volatility clustering (Tsabitah et al., 2025). This behaviour indicates heteroskedasticity in the series, supporting the use of a SARIMA–GARCH model to adequately capture the time-varying variance structure (Mualifah et al., 2024).

SARIMA-GARCH Analysis

1) Testing for Stationarity in Mean and Variance

The stationarity of the variance was first examined using the Box–Cox plot. The initial results showed that the confidence interval for the optimal transformation parameter λ ranged from 2.064 to 3.080. Because this interval did not include the value 1, the series was identified as non-stationary in variance (Purwa & Ngwarati, 2022). To address this issue, a Box–Cox transformation was applied using the optimal λ value of 2.566. A re-evaluation after the transformation indicated that the new λ confidence interval successfully included the value 1, confirming that the variance stationarity problem had been resolved.

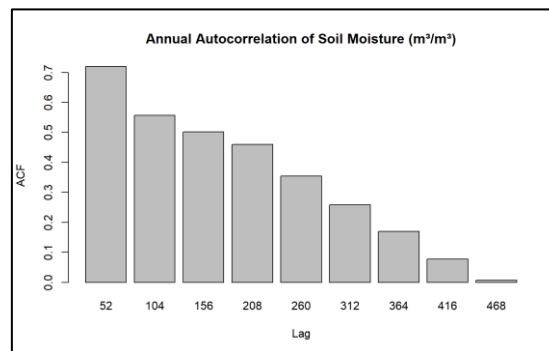


Figure 3. Plot ACF data seasonal

Following the variance stabilisation, the stationarity of the mean was assessed using the Augmented Dickey–Fuller test, which produced a p-value of 0.01 ($p < 0.05$). This result indicates that the transformed series is stationary in its non-seasonal component. However, the ACF (Autocorrelation Function) plot in Figure 3 showed a slow decay at the seasonal lags, suggesting the presence of seasonal non-stationarity. Consequently, seasonal differencing was applied once to address this remaining seasonal pattern.

2) Tentative Model

Based on the ACF, PACF, and EACF plots, several SARIMA candidate models were identified, including those with non-seasonal orders such as (3,0,0), (2,0,2), (3,0,2), and (3,0,5). Furthermore, the seasonal ACF and PACF plots confirmed the seasonal orders to be $P=0$ and $Q=1$. These candidate models will subsequently be evaluated for their goodness-of-fit using various selection indicators.

Table 2. Tentative SARIMA model

Model Candidate	AIC	BIC
ARIMA(3,0,0)(0,1,1)[52]	-2687.7864	-2667.6090
ARIMA(2,0,2)(0,1,1)[52]	-2692.9972	-2668.7843
ARIMA(3,0,2)(0,1,1)[52]	-2689.1919	-2660.9435
ARIMA(3,0,5)(0,1,1)[52]	-2701.0465	-2660.6916

The SARIMA(2,0,2)(0,1,1)[52] model order was selected because it yielded the lowest AIC (-2692.9972) and BIC (-2668.7843) values among all tentative models. To ensure the model's adequacy and parsimony, a diagnostic check was performed by testing models with neighbouring orders; however, the results did not yield a significant improvement in performance. Therefore, the SARIMA(2,0,2)(0,1,1)[52] model was designated as the optimal model.

3) Residual assumption test

Subsequently, assumption testing was performed on the model, and the results are summarized in Table 2.

Table 3. Diagnostic Checking Results of the SARIMA(2,0,2)(0,1,1)[52] Model

Formal Test	Test Method	p-value	Interpretation
Normality Assumption	Shapiro-Wilk test	$5,217 \times 10^{-9}$	The residuals do not follow a normal distribution
No Autocorrelation	Ljung-Box test	$5,510 \times 10^{-1}$	The residuals are independent
Homoscedasticity	Ljung-Box test	$1,799 \times 10^{-14}$	The residuals exhibit heteroscedasticity
Zero-Mean Assumption	t test	$8,123 \times 10^{-2}$	The mean of the residuals is zero

The SARIMA(2,0,2)(0,1,1)[52] model satisfied all diagnostic assumption tests, with the exception of the homoscedasticity assumption (constant variance). This indicated the presence of heteroscedasticity or an ARCH effect. Therefore, to address this variance instability, the SARIMA model was combined with the GARCH method, resulting in the hybrid SARIMA-GARCH model.

4) Overfitting

To verify that the selected ARIMA(2,0,2)(0,1,1)[52] specification was not overfitted, several expanded models were estimated by increasing the non-seasonal (p, q) and seasonal (P, Q) orders. This procedure aimed to assess whether a more complex structure could provide improvements in model quality, as measured by the AIC and BIC.

Table 2. Overfitting model

Model Candidate	AIC	BIC
ARIMA(2,0,3)(0,1,1)[52]	-2697.03	-2668.78
ARIMA(2,0,2)(1,1,1)[52]	-2685.85	-2657.6
ARIMA(2,0,2)(0,1,2)[52]	-2691.13	-2662.88

The results show that none of the expanded models outperforms the original ARIMA(2,0,2)(0,1,1)[52] in terms of both AIC and BIC. Although some alternatives exhibit comparable AIC values, their BIC scores are consistently higher, indicating inferior parsimony. Combined with the statistical significance of the baseline model's parameters, these findings confirm that ARIMA(2,0,2)(0,1,1)[52] remains the most appropriate specification, with no evidence of overfitting.

5) GARCH

The diagnostic checking results presented in Table 4 demonstrate that the SARIMA model successfully satisfies the assumptions of serial independence and zero-mean residuals, as indicated by

the non-significant p -values for the relevant test statistics. However, the assumption of homoscedasticity is clearly violated, with formal tests confirming the presence of significant heteroscedasticity. This lack of variance stability suggests that the standard SARIMA model is inadequate to capture the underlying volatility of the data. Consequently, this necessitates the application of the ARCH-LM test to formally detect autoregressive conditional heteroscedasticity, justifying the subsequent transition to GARCH modelling to better address these dynamic variance patterns.

Table 4. ARCH-LM test result

Lag	P-value
1	$1,4203 \times 10^{-14*}$
2	$1,4203 \times 10^{-14*}$
...	...
...	...
51	$1,4203 \times 10^{-14*}$
52	$1,4203 \times 10^{-14*}$

*Significant at the 5% level

The ARCH-LM test results indicate that all p -values up to lag 52 are statistically significant, thus confirming the presence of heteroscedasticity in the residuals. Consequently, the assumption of residual homogeneity in the SARIMA model is violated. The significance of these high-order lags suggests a persistent volatility effect or long-term dependency in the variance. To address this issue and capture the dynamic behaviour of the variance, further modelling using the GARCH framework is required (Li et al., 2021).

Table 5. Results of tentative model identification.

Model	Parameter Significance	AIC
GARCH(1,1) NORM	α_1, β_1	-7.1159
GARCH(1,1) T-STUD	α_1, β_1 , shape	-7.4957
GARCH(1,2) NORM	α_1, β_1	-7.1060
GARCH(1,2) T-STUD	shape	-7.5114
GARCH(2,1) NORM	α_1, β_1	-7.1066
GARCH(2,1) T-STUD	α_1, β_1 , shape	-7.4697
GARCH(2,2) NORM	$\alpha_1, \alpha_2, \beta_2$	7.1049
GARCH(2,2) T-STUD	$\alpha_1, \alpha_2, \beta_1$, shape	-7.5098

Based on the model selection criterion of the lowest AIC and the significance of the estimated parameters, the GARCH(1,1) model with a Student-t error distribution was selected as the most appropriate specification. Model validation using the Ljung–Box test on the squared residuals produced non-significant p -values, indicating that no autocorrelation remains in the squared residuals and that the heteroscedasticity issue has been resolved. Following this, additional diagnostic tests were performed. The Ljung–Box test on the fitted residuals indicated white-noise behaviour, while the Jarque–Bera test remained significant, showing that the residuals are not normally distributed. However, the ARCH LM test was non-significant, confirming that no remaining ARCH effects persist. Overall, these results indicate that the combined SARIMA(2,0,2)(0,1,1)[52] – GARCH(1,1) model is the best-fitting model.

LSTM Analysis

The LSTM modelling procedure is carried out through two principal stages: the data transformation phase and the subsequent hyperparameter tuning process.

1) Data Transformation

The initial phase of LSTM modelling involves data transformation to ensure input stability. Specifically, StandardScaler is employed to standardise the data by transforming each value into its corresponding *z-score*.

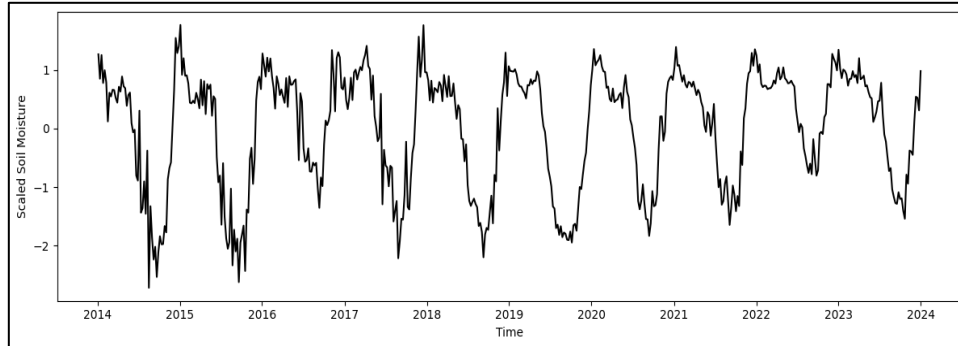


Figure 4. Data after standard scaled transformation.

Figure 3 illustrates the soil moisture time series after undergoing a standardization process. This procedure rescales the data to have a mean of 0 and a standard deviation of 1, thereby reducing the influence of varying feature magnitudes and facilitating more stable and efficient training of the LSTM network.

2) Hyperparameter Tuning

This study employs a manual grid search strategy to optimize batch size, epochs, and learning rate. Under this approach, various parameter combinations are systematically tested via manual iteration. The performance of each combination is evaluated, and the specific setup resulting in the minimum loss is chosen as the final optimal configuration.

Table 6. Hyperparameter tuning

Model	Batch Size	Epoch	Drop Out	Test RMSE
1	16	50	0.1	0.052198
2			0.05	0.020113
3		100	0.1	0.016683
4			0.05	0.030915
5	32	50	0.1	0.016919
6			0.05	0.016439
7		100	0.1	0.015791
8			0.05	0.023672
9	64	50	0.1	0.024604
10			0.05	0.016101
11		100	0.1	0.034136
12			0.05	0.023407

The hyperparameter tuning results identifying the optimal configuration for the proposed LSTM network are presented in Table 6. The best-performing setting corresponds to 100 epochs, a batch size of 32, and a dropout rate of 0.10, which together produced the lowest Test RMSE of 0.015791. An epoch count of 100 indicates that the network completed 100 full passes over the training data, allowing repeated weight adjustments to refine pattern learning. The batch size of 32 means that

gradients were computed using groups of 32 samples per iteration, balancing computational efficiency and learning stability. Meanwhile, the dropout rate of 0.10 reflects the random deactivation of 10% of neurons during training, a mechanism intended to reduce overfitting and improve generalization performance.

Model Evaluation

The final evaluation phase compares the forecasting accuracy of the hybrid statistical method, SARIMA-GARCH, against the Deep Learning LSTM approach using test data from January to December 2023. The performance metrics employed include RMSE and MAPE to assess the deviation of the forecasted values from the actual observed data. A summary of the comparative evaluation results for both methods is presented in Table 6.

Table 7. Model Evaluation		
Model	RMSE Test	MAPE Test
SARIMA-GARCH	0.0399	9.94
LSTM	0.0157	3.52

Based on the results in Table 7, the LSTM method demonstrates clear superiority over the SARIMA-GARCH model across both evaluation metrics. This advantage is especially evident in the RMSE performance, where the LSTM achieved a value of 0.0157, substantially lower than the 0.0399 recorded by SARIMA-GARCH. A similar pattern is observed in the MAPE results: the LSTM model produced an error of 3.52, which is markedly lower than the 9.94 obtained by SARIMA-GARCH. These findings indicate that although the SARIMA-GARCH framework is capable of addressing seasonality and volatility, the LSTM model is considerably more effective in capturing the nonlinear and complex temporal dynamics present in soil moisture data.

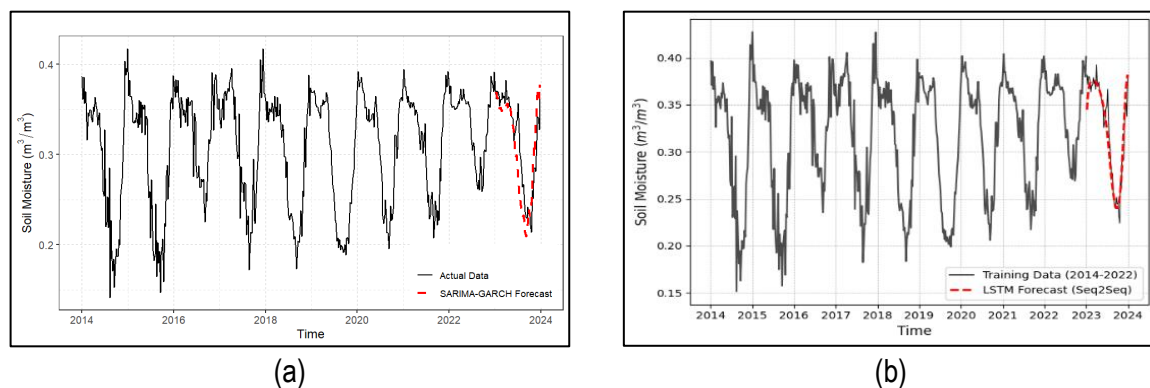


Figure 5. Comparison between the actual data and the forecasting results on the test data (January–December 2023): (a) SARIMA-GARCH; (b) LSTM

Figures 5 demonstrate that while both models correctly capture the seasonal timing of soil moisture, the LSTM model outperforms the SARIMA-GARCH model in terms of stability and realism. The SARIMA-GARCH forecast is characterized by excessive noise and jagged fluctuations, indicating sensitivity to residual variance. In contrast, the LSTM model produces a smooth and continuous trajectory that aligns more naturally with the physical inertia of the data. This visual evidence suggests that the LSTM's deep learning architecture is superior in handling the non-linear dependencies of the time series compared to the statistical approach.

CONCLUSION

This study concludes that the LSTM model is significantly superior to the SARIMA-GARCH model for forecasting soil moisture in Indramayu Regency. While SARIMA-GARCH adequately addresses volatility, it yields higher error rates and noisy predictions. In contrast, the LSTM model demonstrated exceptional accuracy, achieving an RMSE of 0.0157 and MAPE of 3.52%, compared to 9.94% for the statistical benchmark. Furthermore, the LSTM forecasts exhibited a stable and realistic seasonal trajectory. Therefore, the LSTM architecture is recommended as the most effective tool for supporting irrigation management and agricultural decision-making in the region.

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